

OMKAR MAHESH KASHYAP

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EDUCATION

PES University

Bachelor of Technology in Computer Science & Engineering, **GPA 9.35/10**

Aug 2022 - Apr 2026

Bengaluru, India

RV PU College

Grade XII, Karnataka State Board, **Percentage: 98.3%, State Rank 9**

Jun 2020 - Jun 2022

Bengaluru, India

RESEARCH & WORK EXPERIENCE

Vision and AI Lab, Indian Institute of Science

Research Intern

Advisor: Prof. R Venkatesh Babu

Oct 2025 - Present

Bengaluru, India

- Developed an unsupervised hierarchical segmentation algorithm to automatically identify and group semantically coherent objects within complex visual scenes, enhancing scene understanding through structured decomposition.
- Proposed an object-centric representation learning framework for vision-language models that replaces conventional image patches with detected object regions, strengthening multimodal alignment and improving object-level comprehension and reasoning.
- Trained and benchmarked multiple vision-language architectures on diverse evaluation suites, including POPE and CHAIR, achieving 50% faster inference and a 25% reduction in hallucination rates.

University of California, Berkeley

Research Intern

Advisor: Prof. Prasad Raghavendra

Jun 2025 - Present

Berkeley, CA

- Investigated whether sparse neural networks trained using gradient descent combined with L1 regularization can learn structured sparse linear transforms, such as Hadamard and Fourier transforms. This involved mechanistic interpretability experiments to study how these networks internally represent and approximate known sparse algorithms.
- Designed and trained sparse neural network architectures, analyzing the learned weight connectivity patterns and their evolution to reveal emergent computational circuits that efficiently implement the sparse transforms under study.

Center for Data Sciences and Applied Machine Learning, PES University

Machine Learning Research Intern

Advisor: Prof. Shylaja SS, Prof. Ashwini M Joshi

Jun 2024 - Aug 2024

Bengaluru, India

- Conducted research on hypergraph-based models for Aspect-Based Sentiment Analysis (ABSA), focusing on short-text scenarios with limited data, addressing challenges in capturing complex aspect-sentiment relationships.
- Co-developed a novel hypergraph construction technique utilizing hierarchical clustering with adaptive thresholding, which dynamically determines the granularity of aspect-opinion groupings, improving the model's ability to capture nuanced aspect-sentiment dependencies.
- Led extensive experiments and ablation studies, achieving a state-of-the-art F1-Score of 0.83 on benchmark datasets and validating the effectiveness of the hypergraph approach against multiple graph-based baselines.

PUBLICATIONS

(* INDICATES EQUAL CONTRIBUTION)

[1] From Graphs to Hypergraphs: Enhancing Aspect-Based Sentiment Analysis via Multi-Level Relational Modeling

Omkar Mahesh Kashyap*, Padegal Amit*, Madhav Kashyap, Ashwini M. Joshi, Shylaja S. S.

arXiv preprint, 2025.

[PDF]

[2] Quantifying Modality Contributions via Disentangling Multimodal Representations

Padegal Amit*, Omkar Mahesh Kashyap*, Namitha Rayasam*, Nidhi Shekhar, Surabhi Narayan

arXiv preprint, 2025.

[PDF]

AWARDS & GRANTS

- Junior Research Fellowship:** PES University-funded grant awarded to support independent undergraduate research **Jan 2025**
- MRD Merit Scholarship:** 25% tuition fee waiver to top 1% students at PES University **Dec 2022, May 2023, Dec 2024**
- CNR Rao Scholarship:** 20% tuition fee waiver to top 20% students at PES University **May 2025, May 2024, Dec 2023**
- State Rank 9,** Karnataka Pre-University (12th Grade) Board Examination **Jun 2022**
- Carnatic Classical Vocal Music:** Junior Certification **May 2016**

LEADERSHIP & COMMUNITY INVOLVEMENT

AI/ML Core Member, Google Developer Student Clubs, PES University

Aug 2023 - Present

Organized an open-source contribution workshop to foster community engagement and collaborative learning, contributing hands-on by implementing a YOLOv8-based object detection system for real-time video streams, and earned recognition by winning a Kaggle competition among 200+ participants.

Mentor, Kodikon 5.0 Nationwide Hackathon, India

Oct 2025

Developed problem statements and mentored student teams by providing structured guidance throughout their projects, helping teams define objectives, plan approaches, and implement solutions.

TECHNICAL SKILLS

- **Languages:** Python, C++, JavaScript, HTML, SQL
- **Technologies/Frameworks:** PyTorch, Kubernetes, Docker, Spark, OpenCV, NumPy, Pandas, Linux, Git, AWS
- **Relevant Courses:** Machine Learning, Image Processing and Computer Vision, Large Language Models, Data Structures and Algorithms, Operating Systems, Computer Networks, Database Management Systems, Cloud Computing, Heterogeneous Parallelism

PROJECTS

Virtual TryOn

- Engineered a Virtual Try-On web application using React and Flask, integrating TensorFlow for image processing, and leveraging a NoSQL backend for efficient data management.
- Constructed a real-time image processing pipeline capable of handling 30+ frames per second with sub-200ms latency by optimizing TensorFlow models and implementing efficient NoSQL caching strategies.
- Delivered a cross-platform virtual fitting experience with 90% user satisfaction ratings by developing custom overlay techniques and responsive design principles. Open-sourced at [GitHub link](#).

Knowledge Graph Generator

- Built a full-stack web app to convert user-input text into interactive knowledge graphs using a locally hosted open-source LLM and Neo4j for dynamic entity-relation visualization.
- Engineered a modular backend with FastAPI and Docker to handle text processing, graph management, and API orchestration; leveraged Hugging Face Transformers for summary and entity extraction.
- Designed an intuitive frontend using Next.js, Cytoscape.js, and Zustand for state management, enabling real-time graph rendering and user-driven knowledge expansion. Open-sourced at [GitHub link](#).

Distributed Systems Simulator

- Developed a lightweight distributed-systems simulator in Python using Flask and packaged it in a Docker container to emulate core Kubernetes control-plane operations.
- Established a complete RESTful API ecosystem processing 100+ operations per second with average latency under 50ms by implementing efficient request handling and asynchronous task queuing.
- Integrated a volume-mounted SQLite database to persist cluster metadata across container restarts. Open-sourced at [GitHub link](#).